Aquifers, reservoirs, wetlands

SWAT+ Documentation

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Purpose

The aquifer group has two input files declared in `type input_aqu` (`src/input_file_module.f90`):

- `aquifer.aqu` defines one record per aquifer object. Each record sets baseflow recession parameters, initial water table depth, initial chemistry, and the recharge and revap controls used by the lumped (non-gwflow) aquifer module. Records are read into `aqudb` (an array of `type aquifer_database`).
- `initial.aqu` is a small table of named initial-condition pointers. Each entry points to organic-mineral, pesticide, pathogen, heavy-metal, and salt initialization files. The `init` column of each row in `aquifer.aqu` selects one of these names.

Source

- Readers: `src/aqu_read.f90`, `src/aqu_read_init.f90`
- Type definitions: `src/aquifer_module.f90`, `type aquifer_database` and `type aquifer_init_data_char`
- Filename declarations: `src/input_file_module.f90`, `type input_aqu`

aquifer.aqu

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per aquifer with list-directed I/O. The first integer on each record is the aquifer id; the reader uses it as the index into `aqudb`. The remaining columns are read into the fields of `type aquifer_database` in declaration order.

#	Field	Type	Units	Default	Description
1	id	int	none	-	aquifer id. Used as the array index into `aqudb`
2	aqunm	char(16)	none	“”	aquifer name
3	aqu_ini	char(16)	none	“”	initial-condition pointer. Cross-walks with `name` in `initial.aqu`
4	flo	real	mm	0.05	flow from aquifer in the current time step
5	dep_bot	real	m	0.0	depth from mid-slope surface to bottom of aquifer
6	dep_wt	real	m	0.0	initial depth from mid-slope surface to water table
7	no3	real	ppm N	0.0	initial nitrate-N concentration in aquifer
8	minp	real	ppm P	0.0	initial mineral phosphorus concentration in aquifer
9	cbn	real	percent	0.5	initial organic carbon in aquifer
10	flo_dist	real	m	50.0	average flow distance to stream or downstream object
11	bf_max	real	mm	0.0	maximum daily baseflow when all channels are contributing
12	alpha	real	1/day	0.0	lag factor for the groundwater recession curve
13	revap_co	real	none	0.0	revap coefficient. `evap = pet * revap_co`
14	seep	real	fraction	0.0	fraction of recharge that seeps from the aquifer to the deep store
15	spyld	real	$m^{3/m}3$	0.0	specific yield of the aquifer
16	hlife_n	real	days	30.0	half-life of nitrogen in groundwater
17	flo_min	real	m	0.0	water table depth above which return flow occurs
18	revap_min	real	m	0.0	water table depth above which revap occurs

Example

`refdata/Osu_1hru/aquifer.aqu`:

```
aquifer.aqu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                          init        gw_flo       dep_bot        dep_wt         no3_n
       1  aqu011                     initaqu1       0.05000      10.00000       6.00000       0.00000
       2  aqu012                     initaqu1       0.05000      10.00000       6.00000       0.00000
      21  aqu_deep010                initaqu1       0.00000      10.00000      20.00000       0.00000
```

Notes:

- The tool header names (`gw_flo`, `no3_n`, `sol_p`, `carbon`, `alpha_bf`, `revap`, `rchg_dp`, `spec_yld`, `hl_no3n`) differ from the source field names (`flo`, `no3`, `minp`, `cbn`, `alpha`, `revap_co`, `seep`, `spyld`, `hlife_n`). Column order is what matters.
- The `init` column points to `initaqu1`, which must exist as a row in `initial.aqu`.
- In this dataset all shallow aquifers share the same property values. The last record (`aqu_deep010`) represents the deep aquifer with no baseflow (`flo = 0`, `flo_min = 0`, `revap_min = 0`).

initial.aqu

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per initial-condition set with list-directed I/O. Records are read into `aqu_init_dat_c` (an array of `type aquifer_init_data_char`).

#	Field	Type	Description
1	name	char(16)	initial set name. Cross-walks with `aqu_ini` in `aquifer.aqu`
2	org_min	char(16)	name of organic-mineral initial-state record
3	pest	char(16)	name of pesticide initial-state record
4	path	char(16)	name of pathogen initial-state record
5	hmet	char(16)	name of heavy-metal initial-state record
6	salt	char(16)	name of salt initial-state record

The literal `null` is accepted in slots that have no initial-state file.

Example

`refdata/Osu_1hru/initial.aqu`:

```
initial.aqu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                       org_min             pest             path             hmet             salt
initaqu1                   no_init             null             null             null             null
```

Notes:

- The tool writes a trailing `description` column. The source type has no `description` field. List-directed reads consume tokens left to right, so the extra trailing token is harmless. Do not rely on the description being read.
- `no_init` is itself a name that must resolve in the organic-mineral initial file. `null` is treated as "no file".

Related files

- aquifer.con. Connectivity entries that map each aquifer id to its spatial location and downstream object.
- `aqu_catunit.ele` and `aqu_catunit.def`. Catchment-unit elements that group aquifers.
- Organic-mineral, pesticide, pathogen, heavy-metal, and salt initial-state files referenced by `initial.aqu`.

Purpose

The reservoir group has six input files declared in `type input_res` (`src/input_file_module.f90`):

- `reservoir.res` is the object table. One record per reservoir, naming the hydrology, sediment, nutrient, release, and initial-condition records that apply.
- `initial.res` is a table of named initial-condition pointers (organic-mineral, pesticide, pathogen, heavy-metal, salt) shared with the wetland module.
- `hydrology.res` is the hydrology database. One record per named hydrology profile.
- `sediment.res` is the sediment-quality database.
- `nutrients.res` is the nutrient settling and loss-rate database.
- `weir.res` is the weir outflow database for reservoirs that release through a weir.

All five database files are referenced by name from `reservoir.res`. Names are not positional. The reader builds an index after all files are loaded.

Source

- Readers:
  - src/res_read.f90
  - src/res_read_init.f90
  - src/res_read_hyd.f90
  - src/res_read_sed.f90
  - src/res_read_nut.f90
  - src/res_read_weir.f90
- Type definitions: src/reservoir_data_module.f90
- Filename declarations: src/input_file_module.f90, `type input_res`

reservoir.res

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per reservoir with list-directed I/O. The first integer on each record is the reservoir id; the reader uses it as the index into `res_dat_c`. The remaining columns are read into `type reservoir_data_char_input` in declaration order.

#	Field	Type	Description
1	id	int	reservoir id. Used as the array index into `res_dat_c`
2	name	char(25)	reservoir name
3	init	char(25)	initial-state pointer. Cross-walks with `init` in `initial.res`
4	hyd	char(25)	hydrology pointer. Cross-walks with `name` in `hydrology.res`
5	release	char(25)	release rule pointer. Either a decision-table name in `res_rel.dtl`, or a name prefixed with `ctbl_` for a conditional table
6	sed	char(25)	sediment pointer. Cross-walks with `name` in `sediment.res`
7	nut	char(25)	nutrient pointer. Cross-walks with `name` in `nutrients.res`

The `release` lookup branches on the literal prefix `ctbl_`. If present, the name is matched against `ctbl(:)%name` and the release mode is set to `c`. Otherwise it is matched against `dtbl_res(:)%name` and the release mode is set to `d`.

initial.res

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per initial-condition set with list-directed I/O. Records are read into `res_init_dat_c` (an array of `type reservoir_init_data_char`).

#	Field	Type	Description
1	init	char(25)	initial set name. Cross-walks with `init` in `reservoir.res` (and `wetland.wet`)
2	org_min	char(25)	name of organic-mineral initial-state record
3	pest	char(25)	name of pesticide initial-state record
4	path	char(25)	name of pathogen initial-state record
5	hmet	char(25)	name of heavy-metal initial-state record
6	salt	char(25)	name of salt initial-state record

The literal `null` is accepted in slots that have no initial-state file.

Example

`refdata/Osu_1hru/initial.res`:

```
initial.res: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                      org_min             pest             path             hmet             salt
initwet1                  no_init             null             null             null             null
```

The first column header in the example is `name` but the source field is `init` (the value it must match in `reservoir.res`). The source uses position. The editor also writes a trailing `description` column that is not read.

hydrology.res

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per hydrology profile with list-directed I/O. Records are read into `res_hyddb` (an array of `type reservoir_hyd_data`) in declaration order.

#	Field	Type	Units	Default	Description
1	name	char(25)	none	“default”	hydrology profile name
2	iyres	int	none	0	simulation year the reservoir becomes operational
3	mores	int	none	0	month the reservoir becomes operational
4	psa	real	ha	0.0	surface area when filled to principal spillway
5	pvol	real	ha-m	0.0	volume needed to fill to principal spillway. Converted to m^3 by `res_initial`
6	esa	real	ha	0.0	surface area when filled to emergency spillway
7	evol	real	ha-m	0.0	volume needed to fill to emergency spillway. Converted to m^3 by `res_initial`
8	k	real	mm/hr	0.01	hydraulic conductivity of the reservoir bottom
9	evrsv	real	none	0.7	lake evaporation coefficient
10	br1	real	none	0.0	volume-surface area coefficient (estimated by the model if zero)
11	br2	real	none	0.0	volume-surface area coefficient (estimated by the model if zero)

The reader fills in defaults when fields are zero or non-positive:

- If `pvol + evol > 0` and `pvol <= 0`, sets `pvol = 0.9 * evol`.
- If both are zero or negative, sets `pvol = 60000.0`.
- If `evol <= 0`, sets `evol = 1.11 * pvol`.
- If `psa <= 0`, sets `psa = 0.08 * pvol`.
- If `esa <= 0`, sets `esa = 1.5 * psa`.
- If `evrsv <= 0`, sets `evrsv = 0.6`.

Note the default for `evrsv` in the type definition is `0.7`, but the runtime default applied when the field is zero or negative is `0.6`.

sediment.res

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per sediment profile with list-directed I/O. Records are read into `res_sed` (an array of `type reservoir_sed_data`).

#	Field	Type	Units	Description
1	name	char(25)	none	sediment profile name
2	nsed	real	mg/L	normal amount of sediment in the reservoir
3	d50	real	um	median particle size of suspended and benthic sediment
4	carbon	real	percent	organic carbon in suspended and benthic sediment
5	bd	real	t/m^3	bulk density of benthic sediment
6	sed_stlr	real	none	sediment settling rate
7	velsetlr	real	m/day	sediment settling velocity

Example

`refdata/Osu_1hru/sediment.res`:

```
sediment.res: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    sed_amt           d50        carbon            bd       sed_stl       stl_vel
sedwet1                   10.00           5.00       0.04            0.8          0.15        0.0002
```

The editor writes `sed_amt`, `sed_stl`, `stl_vel`; the source fields are `nsed`, `sed_stlr`, `velsetlr`. Column order is what matters.

nutrients.res

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per nutrient profile with list-directed I/O. Records are read into `res_nut` (an array of `type reservoir_nut_data`). After reading, the reader divides the four `setlr` rates and the two `solr` rates by 365 (m/yr to m/day).

#	Field	Type	Units (file)	Description
1	name	char(25)	none	nutrient profile name
2	ires1	int	day of year	start of mid-year settling season
3	ires2	int	day of year	end of mid-year settling season
4	nsetlr1	real	m/year	nitrogen mass loss rate, mid-year period
5	nsetlr2	real	m/year	nitrogen mass loss rate, remainder of year
6	psetlr1	real	m/year	phosphorus mass loss rate, mid-year period
7	psetlr2	real	m/year	phosphorus mass loss rate, remainder of year
8	nsolr	real	m/year	loss rate for soluble nitrogen (NO3, NH3, NO2)
9	psolr	real	m/year	loss rate for soluble phosphorus
10	theta_n	real	none	temperature adjustment for nitrogen settling

#	Field	Type	Units (file)	Description
11	theta_p	real	none	temperature adjustment for phosphorus settling
12	conc_nmin	real	ppm	minimum nitrogen concentration for settling
13	conc_pmin	real	ppm	minimum phosphorus concentration for settling

Example

`refdata/Osu_1hru/nutrients.res`:

```
nutrients.res: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name               mid_start   mid_end     mid_n_stl         n_stl     mid_p_stl         p_stl       chla_c
nutwet1                    5        10       5.50000       5.50000      10.00000      10.00000       1.00000
```

Notes:

- Editor headers `mid_n_stl`, `n_stl`, `mid_p_stl`, `p_stl` correspond to source fields `nsetlr1`, `nsetlr2`, `psetlr1`, `psetlr2`.
- Editor headers `chla_co` and `secchi_co` correspond to source fields `nsolr` and `psolr` by position. These are the soluble nitrogen and phosphorus loss rates, not chlorophyll or Secchi-depth coefficients.

weir.res

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per weir profile with list-directed I/O. Records are read into `res_weir` (an array of `type reservoir_weir_outflow`).

#	Field	Type	Units	Default	Description
1	name	char(25)	none	“”	weir profile name
2	c	real	none	1.84	weir discharge linear coefficient
3	k	real	none	2.6	weir discharge exponential coefficient
4	w	real	m	2.5	weir width
5	h	real	m	0.0	height of the weir above the bottom of the impoundment

Example

`refdata/Osu_1hru/weir.res`:

```
weir.res: Reservoir weir inputs - asdf;lj
NAME   Linear_C    Exp_K     Width(m)   Depth(m)
weir1   1.83       1.50      5.00       0.0
```

Related files

- reservoir.con. Connectivity entries that place each reservoir in the routing topology.
- `res_rel.dtl`. Decision tables referenced from the `release` column of `reservoir.res`.
- wetland.wet. Wetlands share the `initial.res` table and the `hydrology.wet`, `sediment.res`, `nutrients.res`, and `weir.res` databases.

Purpose

The wetland group has two input files declared in `type input_res` (`src/input_file_module.f90`):

- `wetland.wet` is the wetland object table. One record per wetland, naming the initial-condition set, hydrology profile, release rule, sediment profile, nutrient profile, and weir profile that apply.
- `hydrology.wet` is the wetland hydrology database. One record per named hydrology profile. The geometry is HRU-relative (fraction of HRU area, depth in mm) rather than absolute area and volume as for reservoirs.

Wetlands reuse the reservoir sediment, nutrient, weir, and initial-condition databases. The wetland records in `wetland.wet` point at the same `sediment.res`, `nutrients.res`, `weir.res`, and `initial.res` tables described in reservoir.res.

Source

- Readers: src/wet_read.f90, src/wet_read_hyd.f90
- Type definitions: src/reservoir_data_module.f90, `type reservoir_data_char_input` (shared with reservoirs) and `type wetland_hyd_data`
- Filename declarations: src/input_file_module.f90, `type input_res`

wetland.wet

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per wetland with list-directed I/O. The first integer on each record is the wetland id (used as the index into `wet_dat_c`). The remaining columns are read into `type reservoir_data_char_input` in declaration order (the same type used by reservoirs).

#	Field	Type	Description
1	id	int	wetland id. Used as the array index into `wet_dat_c`
2	name	char(25)	wetland name
3	init	char(25)	initial-state pointer. Cross-walks with `init` in `initial.res`
4	hyd	char(25)	hydrology pointer. Cross-walks with `name` in `hydrology.wet`
5	release	char(25)	release rule pointer (decision-table name)
6	sed	char(25)	sediment pointer. Cross-walks with `name` in `sediment.res`
7	nut	char(25)	nutrient pointer. Cross-walks with `name` in `nutrients.res`

The example dataset writes a trailing `WEIR` column. The base type has six character fields (name, init, hyd, release, sed, nut). A weir pointer is held in the extension type `reservoir_data` (`character (len=25) :: weir`). The list-directed reader for `wetland.wet` does not consume a separate weir token into the base record; the source links wetland weirs through dedicated reservoir-conditions input. See the Important note below.

Example

`refdata/Osu_1hru/wetland.wet`:

```
wetland.       wet: written by SWAT+ edi#############for SWAT+ rev.60.5.4
     id                name         init          hyd          rel          sed          nut         WEIR
       1           paddy0001     initwet2        paddy         weir      sedwet1      nutwet1        weir1
```

Notes:

- The title line in this dataset is partially corrupted ("wetland. wet:"). The reader skips it as a single text line, so the content does not matter as long as it stays on one line.
- The `rel` column value `weir` is matched against decision-table names. If no match exists in `res_rel.dtl` (or in the `ctbl_` conditional tables), the wetland will be flagged as missing a release rule.

hydrology.wet

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per hydrology profile with list-directed I/O. Records are read into `wet_hyddb` (an array of `type wetland_hyd_data`) in declaration order.

#	Field	Type	Units	Default	Description
1	name	char(25)	none	“default”	hydrology profile name
2	psa	real	fraction	0.0	fraction of HRU area at the principal spillway. Surface-inlet riser flow starts at this level
3	pdep	real	mm	0.0	average water depth at the principal spillway
4	esa	real	fraction	0.0	fraction of HRU area at the emergency spillway. Overflow into the ditch starts at this level
5	edep	real	mm	0.0	average water depth at the emergency spillway
6	k	real	mm/hr	0.01	hydraulic conductivity of the wetland bottom
7	evrsv	real	none	0.7	wetland evaporation coefficient
8	acoef	real	none	1.0	volume-surface area coefficient for HRU impoundment
9	bcoef	real	none	1.0	volume-depth coefficient for HRU impoundment
10	ccoef	real	none	1.0	volume-depth coefficient for HRU impoundment
11	frac	real	none	0.5	fraction of HRU that drains into the impoundment

The reader fills defaults when fields are zero or non-positive:

- If `psa <= 0`, sets `psa = 0.08 * wet_hyd(ires)%pdep` (uses the previously initialized `wet_hyd`, not the just-read `wet_hyddb`).
- If `esa <= 0`, sets `esa = 1.5 * wet_hyd(ires)%psa`.
- If `evrsv <= 0`, sets `evrsv = 0.6`.

Example

`refdata/Osu_1hru/hydrology.wet`:

```
hydrology.wet: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                     hru_ps         dp_ps        hru_es         dp_es             k          evap    vol
paddy                      1.00        150.00          1.00        150.00           0.5          0.75
```

Editor headers `hru_ps`, `dp_ps`, `hru_es`, `dp_es`, `evap`, `vol_area_co`, `vol_dp_a`, `vol_dp_b`, `hru_frac` correspond to source fields `psa`, `pdep`, `esa`, `edep`, `evrsv`, `acoef`, `bcoef`, `ccoef`, `frac`. Column order is what matters.

Related files

- `reservoir.res` and `initial.res`. Shared tables. `wetland.wet` records point at `initial.res` names, and the sediment, nutrient, and weir lookups use the same databases as reservoirs.
- HRU connectivity. Each wetland is attached to an HRU; the geometry in `hydrology.wet` is expressed as fractions of HRU area.